Thangal Kunju Musaliar College of Engineering
(Government Aided and Autonomous)

Kollam, Kerala

IDEA LAB PROJECT REPORT

Department of Mechanical Engineering

May 2025

https://sites.google.com/view/waterqualitytester/home

Thangal Kunju Musaliar College of Engineering

(Government Aided and Autonomous)

IDEA LAB PROJECT REPORT

CERTIFICATE

Certified that this is a bonafide record of the group project work done by ANANTHAKRISHNAN SUBASHBABU, MANIKANDAN G, SALWAN S ZAIN of Semester 2, Roll no B24MEA26, B24MEA43, B24MEA59 for IDEA Lab during the year 2024 - 2025.

VISION

Excellence in mechanical engineering with a perspective of sustainable development.

MISSION

Promote effective teaching-learning through a congenial and disciplined environment. Encourage continuous learning and promote collaborative projects and research. Motivate and equip the students to take up real-life engineering problems with ethical, social and environmental outlook.

Promote co-curricular and extra-curricular activities for the overall development of students.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply principles of mathematics, science and engineering to analyse and design thermal and mechanical systems.

PSO2: Use the knowledge of manufacturing science, state-of-the-art tools and sustainable approaches to evolve processes and products.

PSO3: Manage projects by applying principles of engineering management following ethical standards.

PROGRAMME OUTCOMES

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **Conduct investigation of complex problem:** Use research- based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitation.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **Environment and sustainability**: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage project and in multidisciplinary environment.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES (COs)

COI	Develop project using appropriate Microcontroller Programming languages. [Apply level]

C02	Develop product using PCB Design and Prototyping concepts. [Apply level]
C03	Create 2D and 3D models using appropriate tools. [Apply level]
C04	Create electronic documentation for the system/project using appropriate tools. [Apply level]
C05	Build useful and standalone system/ project with enclosures. [Apply level]

SMART WATER QUALITY TESTER

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ACKNOWLEDGEMENT

We extend our heartfelt gratitude to the Almighty for blessing us with the strength and perseverance to complete this project. We thank our Head of Department, faculty members, and project guide for their valuable support and technical guidance throughout this endeavor. We are also thankful to our families and peers for their encouragement and assistance.

ABSTRACT

This project aims to monitor water quality using a TDS (Total Dissolved Solids) sensor interfaced with an Arduino microcontroller and display the readings on a 0.96" OLED screen. The system provides a real-time digital display of TDS levels in parts per million (ppm), helping assess the purity of drinking water. This low-cost and portable solution can be deployed for household and field use, promoting health and safety in water consumption.

INTRODUCTION

Water quality is essential for health and safety. The Total Dissolved Solids level is a key parameter indicating the presence of salts, minerals, and impurities. High TDS values may signify contamination or reduced water quality. In this project, we employ a TDS sensor with an Arduino Uno to measure the TDS level and display the value on an OLED screen in real-time.

MOTIVATION

Given the rising concerns about water pollution and the lack of affordable testing equipment, this project was inspired by the need for a simple, accessible solution for individuals and communities to monitor their drinking water quality. Arduino-based platforms offer a cost-effective and customizable option for prototyping such a device.

DESIGN

System Components:

- Arduino Uno / ATmega328PB
- TDS Sensor Module
- 0.96" OLED Display (I2C)

-Wires, Breadboard - Water Sample

Container Block Diagram:

[Water Sample] --> [TDS Sensor] --> [Arduino] --> [OLED Display] Working Principle:

The TDS sensor measures the electrical conductivity of the water, which correlates with them concentration of dissolved solids. The Arduino reads this data via analog input, converts it into TDS value using a formula, and displays it on the OLED.

IMPLEMENTATION

1. Hardware Setup:

- The TDS sensor is connected to Arduino's analog pin (A0).

- The OLED display is connected via I2C
(SDA to A4, SCL to A5 on Arduino Uno).

- Arduino is powered via USB or a 9V battery.

2. Software and Libraries:

- Arduino IDE was used for development.

- Required libraries: Adafruit_GFX.h, Adafruit_SSD1306.h

3. Code Logic:

- Read analog value from the TDS sensor.

- Convert raw voltage to TDS (ppm).

- Display the TDS value on OLED in realtime.

RESULTS

The system accurately measured TDS levels of various water samples (tap water, filtered water,bottled water). OLED display showed clear and readable ppm values. The readings matched standard portable TDS meters with acceptable accuracy. Portable and compact, usable in home or field testing.

CONCLUSION

This Arduino-based TDS monitoring system proved to be a simple yet effective tool for assessing water quality. The OLED display enhances portability and real-time usability. This project can be an educational tool and a prototype for low-cost commercial devices.

FUTURE SCOPE

1. Add temperature compensation for improved accuracy.
2. Integrate with IoT platforms like Blynk or Firebase for remote monitoring.
3. Include data logging on an SD card.
4. Battery-powered version for field deployment.
5. Add additional sensors (pH, turbidity) for full water quality analysis.

REFERENCES

1. Arduino.cc - Arduino Uno Documentation
2. Gravity TDS Sensor - DFRobot Wiki
3. Adafruit OLED Libraries - Adafruit GitHub
4. Environmental Protection Agency (EPA) -Water Quality Standards